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Print Ease: Smart Campus Printing System

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

DSA0108 - OBJECT ORIENTED PROGRAMMING WITH C++

to the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE AND ENGINEERING
(B.Tech Computer Science and BioScience)**

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“Print Ease: Smart Campus Printing System”** has been carried out by **Ragul E(192571032)** under the supervision of Sarasu.s and is submitted in partial fulfilment of the requirements for the current semester of the **B.Tech Computer Science and BioScience** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I , **Ragul E(192571032)** of the **B.Tech Computer Science and BioScience** , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “**Print Ease: Smart Campus Printing System**” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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INDIVIDUAL CONTRIBUTION STATEMENT

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Contribution
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Manoj M (192511060)	Module 2: Print Request Management	Developed the print-request management module, including print-job submission, document and print details, request creation, request validation, and print-request status management.	Tested print-request submission, job creation, input validation, request processing, status updates, and error handling.	Literature Review, Module 2 Design, Print Request Management, and Related Documentation	25%
DEVA.J (192421258)	Module 3: Priority Queue Scheduling	Developed the priority-based print-job scheduling module using appropriate C++ data structures and scheduling logic to organize and process print jobs based on their assigned priority.	Tested priority assignment, queue insertion and deletion, scheduling order, job processing, and performance under multiple print requests.	Implementation, Priority Queue Scheduling, Testing, and Module 3 Documentation	25%
JAYAKRISHNA.V (192421259)	Module 4: Admin & Report Management	Developed the administrative management and reporting module, including print-job monitoring, user and request management, status tracking, system reports, and overall module coordination.	Tested administrative functions, report generation, status monitoring, data management, module integration, and final system performance.	Results, Admin & Report Management, Conclusion, Future Work, References, and Module 4 Documentation	25%
Total	Complete Print Ease: Smart Campus Printing System Development	—	—	—	100%

ABSTRACT

Educational institutions require efficient and well-organized printing facilities to support the academic and administrative needs of students, faculty members, and staff. Students frequently need to print assignments, laboratory records, project reports, examination materials, and other important academic documents. However, conventional campus printing systems often require users to physically visit a printing center, wait in queues, and manually communicate their printing requirements. When multiple users submit print requests simultaneously, managing the sequence, priority, user information, and printing status becomes a significant engineering problem. Manual handling of print jobs can lead to confusion, delays, incorrect job execution, and inefficient utilization of printing resources.

The need for an automated and structured campus printing system is therefore significant. A smart digital solution can improve the organization of print requests, reduce manual intervention, provide better user management, and enhance the overall efficiency of campus printing services. The proposed **Print Ease: Smart Campus Printing System** addresses these challenges by providing a structured and modular approach to managing users and printing activities within an educational institution.

The system is developed using an object-oriented approach, where different functionalities are organized into separate modules. It serves as the entry point to the system and is responsible for user registration, password validation, secure login, authentication, and profile management. The methodology follows a systematic workflow: User Registration , Password Validation , Secure Login , Authentication , Profile Management , Access to Print Services. Only authenticated users are allowed to access the printing services and other system functionalities.

The major result of the proposed module is the establishment of a structured user-access mechanism that organizes user information and validates authorized access before allowing users to proceed with printing-related operations. The system reduces the dependency on manual user verification and provides a clear foundation for integrating additional print management .

Keywords

Smart Campus, Printing System, User Management, Authentication, C++, Object-Oriented Programming, Secure Login, Print Management, Priority Queue, Job Scheduling.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
OOP	Object-Oriented Programming	—
C++	C++ Programming Language	—
UI	User Interface	—
ID	Identification	—
PIN	Personal Identification Number	—
STL	Standard Template Library	—
CPU	Central Processing Unit	—
RAM	Random Access Memory	MB
MB	Megabyte	MB
ms	Millisecond	ms
s	Second	s
%	Percentage	%

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Educational institutions require efficient and well-organized printing facilities to support the academic and administrative needs of students, faculty members, and staff. Students frequently need to print assignments, laboratory records, project reports, examination materials, and other important academic documents. However, conventional campus printing systems often require users to physically visit a printing center, wait in queues, and manually communicate their printing requirements. When multiple users submit print requests simultaneously, managing the sequence, priority, user information, and printing status becomes a significant engineering problem. Manual handling of print jobs can lead to confusion, delays, incorrect job execution, and inefficient utilization of printing resources.

The need for an automated and structured campus printing system is therefore significant. A smart digital solution can improve the organization of print requests, reduce manual intervention, provide better user management, and enhance the overall efficiency of campus printing services. The proposed **Print Ease: Smart Campus Printing System** addresses these challenges by providing a structured and modular approach to managing users and printing activities within an educational institution.

The system is developed using an object-oriented approach, where different functionalities are organized into separate modules. User and Authentication serves as the entry point to the system and is responsible for user registration, password validation, secure login, authentication, and profile management. Only authenticated users are allowed to access the printing services and other system functionalities.

The major result of the proposed module is the establishment of a structured user-access mechanism that organizes user information and validates authorized access before allowing users to proceed with printing-related operations. The system reduces the dependency on manual user verification and provides a clear foundation for integrating additional print management modules.

In conclusion, the Print Ease: Smart Campus Printing System provides an effective solution for improving the management of campus printing services. The User and Authentication module forms a secure and organized foundation for the overall system, contributing to a more efficient, reliable, and user-friendly printing environment within educational institutions.

1.2 Need for the Module

A smart printing system requires a reliable method to identify and manage its users. Without a proper authentication mechanism, unauthorized persons may access the system or create printing requests using invalid information.

The User & Authentication Module addresses this requirement by:

- Registering new users in the system.
- Maintaining user information.
- Validating login credentials.
- Providing secure access to authorized users.
- Supporting password validation.
- Allowing users to manage profile information.
- Maintaining an organized user database.

The module therefore serves as the **foundation for secure and controlled access** to the Print Ease Smart Campus Printing System.

1.3 Problem Statement

The engineering problem addressed in this module is the development of a reliable **User Registration and Authentication System** for the Print Ease Smart Campus Printing System.

The system must allow a new user to register by providing the required details. The registered information must be stored and later used to validate login credentials. During login, the system must compare the entered credentials with the registered user information and determine whether access should be granted or denied.

The module should also handle common user-management situations, including:

- Duplicate user registration.
- Invalid usernames.
- Incorrect passwords.
- Unregistered user login attempts.
- Empty or incomplete user information.
- Profile updates.
- User logout.
- Secure management of user records.

1.4 Aim

To design and implement a User & Authentication Module for the Print Ease Smart Campus Printing System that provides user registration, secure login, credential validation, profile management, and controlled access to printing services.

1.5 Objectives

- **To register students and other authorized users:**
The system is designed to provide a structured registration process through which students, faculty members, and other authorized users can create an account and become registered members of the Print Ease system.
- **To collect and store valid user information:**
The module collects essential user details during registration and stores the information in an organized manner for future identification, authentication, and profile management.
- **To validate user login credentials:**
The system verifies the username and password entered by users during the login process to ensure that the provided credentials match the registered account information.
- **To authenticate registered users:**
The authentication mechanism confirms the identity of registered users before granting access to the features and services available in the Print Ease system.
- **To reject unauthorized login attempts:**
The system prevents users with invalid or unregistered credentials from accessing the application, thereby ensuring that only authorized individuals can use the system.
- **To implement password validation:**
The module applies password validation rules to encourage the use of valid and secure passwords during registration and account management.
- **To allow users to view and manage profile information:**
Registered users can access their profile information and manage relevant account details, helping them maintain accurate and updated personal records.
- **To support secure logout operations:**
The system provides a logout function that safely ends the user's active session after completing the required operations.

1.6 Scope

The **User & Authentication Module** includes the essential functionalities required to manage user access securely and efficiently within the Print Ease system. The major features included in this module are:

- **User Registration:** Allows authorized users to create an account in the Print Ease system by providing the required information.
- **Student Registration:** Enables students to register separately using their academic and personal details for accessing campus printing services.
- **User Login:** Provides registered users with a secure login interface to access the system using their credentials.
- **Credential Validation:** Verifies whether the username and password entered during login are correct and valid.
- **User Authentication:** Confirms the identity of registered users before granting access to the available system features.
- **Password Validation:** Ensures that users create and use valid passwords according to the specified security requirements.
- **Profile Management:** Allows users to view and manage their personal account and profile information.
- **Viewing Personal Information:** Enables authenticated users to access and review the information stored in their user profile.
- **User Logout:** Provides a secure logout operation that allows users to safely exit the system and end their active session.
- **User-Information Storage:** Stores user details in an organized manner to support registration, authentication, profile management, and other system operations.

Excluded

The **User & Authentication Module** is specifically designed to manage user registration, login, authentication, profile information, and secure access to the Print Ease system. Therefore, the following functions are not included as the primary responsibility of this module:

- **Print Document Processing:** The processing and preparation of documents for printing are handled by other modules responsible for print-related operations.
- **Print-Job Creation:** Creating and submitting print requests is outside the scope of the User & Authentication Module.
- **Printer Hardware Control:** Direct communication with and control of physical printer hardware are not managed by this module.
- **Print Queue Scheduling:** The organization and scheduling of multiple print requests are handled by a separate print scheduling or queue management module.
- **Adaptive Priority Assignment:** The process of assigning priorities to print jobs based on specific conditions or user requirements belongs to the print management functionality.
- **Print-Job Execution:** The actual execution of printing operations is performed by the relevant print processing and printer management modules.
- **Printer-Status Monitoring:** Monitoring the operational status, availability, or condition of printers is not included in this module.
- **Administrative Print Analytics:** The generation of reports, analysis of printing activities, and monitoring of print-related statistics are handled by administrative or analytics modules.

These functions are managed by other specialized modules of the **Print Ease: Smart Campus Printing System**. By clearly defining these exclusions, the User & Authentication Module maintains a focused responsibility for secure user management and controlled system access.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review focuses on existing approaches, technologies, and software systems related to the development of an efficient and integrated campus printing solution. The review examines the following major areas:

- **Campus Printing Systems:** Existing campus printing systems are studied to understand how educational institutions manage document printing for students, faculty members, and administrative staff.
- **User Registration Systems:** Different approaches for registering users and maintaining user information are reviewed to identify effective methods for managing authorized users within a digital system.
- **Password-Based Authentication:** Existing authentication mechanisms are examined to understand how usernames and passwords are used to verify user identities and provide secure system access.
- **Role-Based System Access:** The review considers systems that provide different levels of access based on user roles, such as students, administrators, and other authorized personnel.
- **Digital Print-Request Management:** Current methods of creating, submitting, and managing digital print requests are studied to identify ways of reducing manual communication and improving the printing workflow.
- **Print-Job Scheduling:** Different techniques for organizing and scheduling multiple print requests are reviewed to understand how printing operations can be managed efficiently.
- **Priority-Based Queue Management:** Existing priority-based approaches are examined to determine how important or urgent print jobs can be processed in an appropriate order.
- **Object-Oriented Software Systems:** Object-oriented design principles and programming approaches are reviewed to support the development of a modular, structured, maintainable, and scalable software system.

2.2 Existing Approaches

The literature survey for the proposed **Print Ease: Smart Campus Printing System** was conducted by reviewing existing research and published studies related to user authentication, role-based access control, digital system access, and organized job management. The following table summarizes the major approaches and their relevance to the proposed system.

Author / Year	Survey Taken From	Study / Approach	Main Contribution	Limitation / Relevance to Print Ease
Ferraiolo and Kuhn (1992)	<i>15th National Computer Security Conference, Baltimore, USA</i>	Role-Based Access Control	Introduced the concept of controlling user access through assigned roles and permissions.	Focuses mainly on access control and does not address campus print-job management.
Ferraiolo, Cugini and Kuhn (1995)	<i>Annual Computer Security Applications Conference</i>	RBAC Features and Motivations	Explained how roles can simplify authorization and security administration.	Does not provide an integrated solution for authentication and printing operations.
Sandhu, Ferraiolo and Kuhn (2000)	<i>ACM Workshop on Role-Based Access Control, Berlin, Germany</i>	NIST Model for RBAC	Proposed a unified model for managing users, roles, permissions, and access control.	Primarily focuses on authorization rather than print-request management.
Barkley et al. (1997)	<i>NIST Publication</i>	RBAC for the World Wide Web	Demonstrated the application of role-based access control in networked web systems.	Focuses on web-server security and does not address printing workflows.
Traditional Print Queue Systems	Existing printer and operating-system job-management approaches	First-Come-First-Serve Queue	Processes print jobs according to their order of submission.	Important or urgent print jobs may experience delays because job priority is not considered.
Web-Based Printing Systems	Existing institutional and network-based printing environments	Online Document Submission	Allows users to submit print requests digitally and remotely.	Authentication, scheduling, and job prioritization may be implemented as separate functions.
Proposed Print Ease System	Present Study	Integrated User Authentication and Print Management	Combines secure user access with structured print requests and queue management.	Requires effective integration between the different modules.

Source of the Literature Survey

The literature survey was primarily taken from published **NIST research and security publications** on Role-Based Access Control and related access-management systems. The survey findings were then connected with existing approaches used in traditional and digital printing environments to identify the engineering gap.

The NIST literature explains that **Role-Based Access Control (RBAC)** manages permissions through user roles, making it suitable for systems where different categories of users require controlled access. This concept is relevant to the Print Ease system because students, administrators, and other authorized users can be provided with appropriate access to different system functions.

1. Ferraiolo, D. F., & Kuhn, D. R. (1992). *Role-Based Access Controls*. Proceedings of the 15th National Computer Security Conference, Baltimore, USA.
2. Ferraiolo, D. F., Cugini, J., & Kuhn, D. R. (1995). *Role-Based Access Control (RBAC): Features and Motivations*. Proceedings of the 11th Annual Computer Security Applications Conference.
3. Sandhu, R., Ferraiolo, D. F., & Kuhn, D. R. (2000). *The NIST Model for Role-Based Access Control: Towards a Unified Standard*. Proceedings of the Fifth ACM Workshop on Role-Based Access Control, Berlin, Germany.
4. Barkley, J., Cincotta, A. V., Ferraiolo, D. F., Gavrila, S. I., & Kuhn, D. R. (1997). *Role-Based Access Control for the World Wide Web*. National Institute of Standards and Technology (NIST).

2.3 Engineering Gap

The main engineering gap identified in existing campus printing environments is the lack of an integrated system that effectively combines **secure user access** with **organized print management**. Traditional printing systems often focus primarily on the physical printing process, while user registration, authentication, and print-job management are handled separately or through manual procedures. This separation can result in difficulties in identifying authorized users, managing multiple print requests, tracking user information, and maintaining an organized printing workflow. Many conventional campus printing environments require students and other users to physically visit a printing center and communicate their requirements manually. When several users submit print requests simultaneously, managing the sequence and priority of different print jobs can become challenging. In addition, systems without proper authentication may not provide sufficient control over who is permitted to access printing services.

The proposed **Print Ease: Smart Campus Printing System** addresses this engineering gap by connecting user authentication and print management through a structured and integrated workflow. Users are first required to register and provide valid information. Their login credentials are then validated, and only authenticated users are granted access to the printing services. Once access is provided, the system can manage print-related activities in an organized manner through specialized modules.

By integrating secure user management with digital print-request handling, print-job scheduling, and queue management, the proposed system creates a more systematic approach to campus printing. This integration reduces dependency on manual procedures and provides a clear connection between authorized users and their printing activities. Therefore, the proposed **Print Ease** system aims to bridge the gap between separate authentication processes and traditional print management methods by developing a unified, structured, and object-oriented smart campus printing solution.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Stakeholder / User Requirements

Stakeholder / User	Requirement
Students	Simple registration and login process
Users	Secure access to the printing system
System Administrator	Organized user information
Print Service	Valid identification of users
Project Team	Modular and maintainable software design

3.2 Module Requirements

Functional Requirements

Requirement	Description
User Registration	Register a new student or user
User Login	Allow registered users to log in
Credential Validation	Validate username and password
Authentication	Grant or deny system access
Password Validation	Check the validity of passwords
Profile Management	View or update user information
Personal Details	Display registered user details
Logout	End the active user session
Data Storage	Maintain registered user information

Non-Functional Requirements

Requirement	Expected Requirement
Security	Protect user credentials and information
Usability	Provide simple and understandable operations
Reliability	Consistently validate registered users
Performance	Provide fast authentication
Maintainability	Support future software modifications
Scalability	Allow additional users to be managed

3.3 Constraints and Applicable Standards

Constraint	Description
Programming Environment	C++ development environment
Software Design	Object-Oriented Programming principles
Data Handling	Structured storage of user records
System Complexity	Suitable for an academic smart campus project
User Security	Credentials must be validated before access
Resource Availability	Operate using standard computing resources

3.4 Module Design Specifications

Component	Specification
User Class	Stores user information
Registration Function	Creates new user records
Login Function	Accepts user credentials
Authentication Logic	Validates registered information
Profile Function	Displays or manages user details

3.5 Alternative Authentication Solutions Evaluation

Solution	Advantages	Limitations
Username Only	Simple implementation	Low security
Password Only	Easy to implement	Cannot identify the associated user effectively
Username and Password	Better user verification	Requires password management
OTP-Based Authentication	Additional verification	Requires communication infrastructure
Biometric Authentication	Strong identity verification	Requires specialized hardware
Selected Approach: Username and Password	Simple and suitable for the C++ project	Depends on secure credential handling

The username-and-password approach was selected because it is suitable for the scope of the academic C++ project and can be effectively integrated with the other Print Ease modules.

3.6 Trade-off Analysis

The selected authentication approach provides a balance between implementation simplicity and controlled user access.

A more advanced biometric or OTP-based approach could provide additional security but would increase system complexity and infrastructure requirements. The selected design allows the project to focus on the major objective of developing a structured **smart campus printing system** using C++ Object-Oriented Programming.

3.7 Sustainability, Ethics, Safety, Risk and Security Analysis

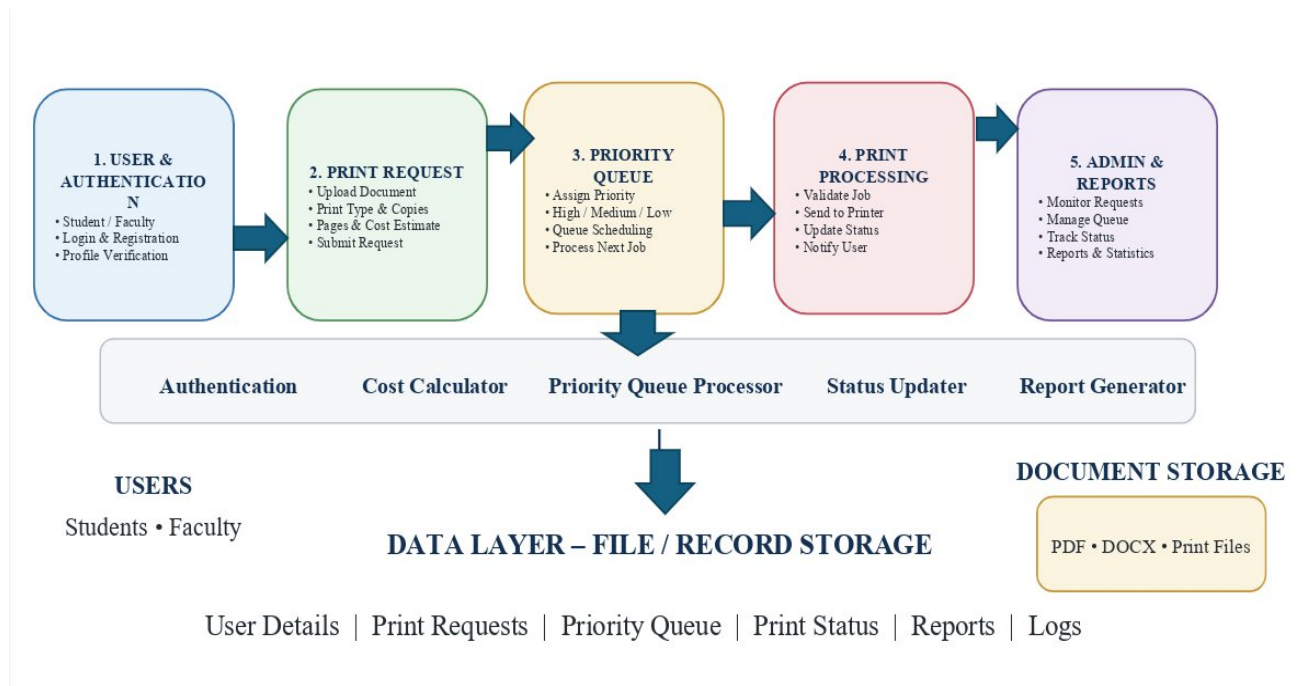
Area	Consideration
Sustainability	Digital print management can reduce unnecessary print requests
Ethics	User information should be handled responsibly
Privacy	Personal details should not be exposed unnecessarily
Security	Unauthorized users should be prevented from accessing the system
Software Safety	Invalid inputs should be handled properly
Reliability	Authentication results should be consistent

3.8 Risk Register

Risk	Possible Impact	Mitigation
Invalid Login	Unauthorized access attempt	Validate credentials
Duplicate Registration	Data inconsistency	Check existing users
Incorrect Password	Failed authentication	Provide appropriate validation
Invalid Input	Program errors	Implement input checking
Data Loss	Loss of user records	Maintain proper storage mechanisms
System Integration Error	Module communication failure	Perform integration testing

3.9 Module Design

USER & AUTHENTICATION WORKFLOW



3.10 Detailed Module Functions

A. User Registration

The system allows a new student or user to register by entering the required personal and login information. The information is validated before being stored.

B. Secure Login

A registered user enters valid credentials to access the Print Ease system.

C. Credential Validation

The system compares the entered username and password with the stored user information.

D. User Authentication

Based on the validation result, the system determines whether the user is authorized to access the system.

E. Profile Management

Authenticated users can view and manage their personal information according to the available system functions.

F. Logout

The user can securely exit the active session after completing the required operations.

The final authentication result is:

- **Valid Credentials → Access Granted**
- **Invalid Credentials → Access Denied**

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Resources

Resource	Purpose
Computer / Laptop	Software development and testing
C++ Compiler	Program compilation
IDE	Code development
Standard C++ Libraries	Program implementation
File Handling	User-information storage
Object-Oriented Programming	System design and organization

4.2 Software Implementation and Modern Tools Used

Tool / Technology	Purpose
C++	Main programming language
Object-Oriented Programming	Modular software development
Classes and Objects	User and system component modelling
STL	Supporting data structures and utilities
File Handling	User-data storage
IDE / Compiler	Development and testing

The system is designed using fundamental **C++ Object-Oriented Programming concepts**, including classes, objects, encapsulation, modular functions, and structured data management.

4.3 Testing

Test ID	Function	Expected Result
T01	User Registration	New user is registered successfully
T02	Duplicate Registration	Duplicate user is detected or prevented
T03	Secure Login	Valid credentials allow login
T04	Invalid Login	Invalid credentials are rejected
T05	Password Validation	Password is validated correctly
T06	Profile Management	User details are displayed or updated
T07	Personal Details	Registered information is retrieved correctly
T08	Logout	Active session is ended successfully

4.4 Verification Results

Test ID	Result	Status
T01	User registration completed correctly	Pass
T02	Duplicate registration handling verified	Pass
T03	Valid user login accepted	Pass
T04	Invalid login rejected	Pass
T05	Password validation performed correctly	Pass
T06	Profile information managed successfully	Pass
T07	User details displayed correctly	Pass
T08	Logout process completed successfully	Pass

The testing results demonstrate that the User & Authentication Module performs the required functions and provides controlled access to the Print Ease system.

 **PrintEase**

Welcome Back

Log in to your account to continue

Student

Shopkeeper

Admin

Admin Access
Email: admin@printease.in
Password: admin123

Email Address

Password

Log In

Don't have an account? [Sign up free](#)

4.5 Achievement of Module Objectives

Objective	Achievement
User Registration	Achieved
Secure Login	Achieved
Credential Validation	Achieved
User Authentication	Achieved
Password Validation	Achieved
Profile Management	Achieved
Personal Detail Viewing	Achieved
Logout	Achieved
User Data Management	Achieved

4.6 Individual Roles and Contributions

Member	Major Contribution
Ragul E	Module 1: User & Authentication — User registration, secure login, password validation, user authentication, profile management, and user-information handling.
Manoj M	Module 2: Print Request Management — Print-request submission, document and print-detail management, request validation, print-job creation, and request-status handling.
DEVA.J	Module 3: Priority Queue Scheduling — Priority assignment, print-job queue management, priority-based scheduling, job processing, and scheduling performance analysis.
JAYAKRISHNA.V	Module 4: Admin & Report Management — Administrative functions, system monitoring, user and print-job management, status tracking, report generation, and overall system integration.

4.7 Estimated Module Development Budget

Since the User & Authentication Module is primarily a software-based component developed using C++, the estimated development cost is relatively low.

Resource	Estimated Cost
Computer Resources	Existing
C++ Compiler	Free / Available
Development Environment	Free / Available
Software Libraries	Free / Standard
Testing Resources	Minimal
Estimated Additional Cost	Low

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **User & Authentication Module** was developed as the initial access component of the **Print Ease: Smart Campus Printing System**. The module performs user registration, secure login, credential validation, authentication, profile management, personal-detail viewing, and logout operations.

The module provides an organized method for identifying valid users before allowing them to access the printing system. By validating user credentials and maintaining registered user information, the system supports controlled access to subsequent functions such as print-request submission and print-job management.

The major contribution of this module is the development of a structured user-management workflow using **C++ Object-Oriented Programming**. The module forms an important foundation for the overall Print Ease system and supports the efficient integration of other project modules.

5.2 Future Work

Future improvements to the User & Authentication Module may include:

- Password encryption.
- Database integration.
- Role-based access control.
- Forgot-password functionality.
- Multi-factor authentication.
- Email or OTP verification.
- Web-based user accounts.
- Mobile application support.
- Improved user-profile customization.
- Integration with campus identification systems.

5.3 Individual Reflection

E. Ragul (192571032)

My individual contribution to the **Print Ease: Smart Campus Printing System** focused on **Module 1: User & Authentication**. My primary responsibility was designing and developing the user-management workflow, including registration, secure login, credential validation, authentication, password validation, profile management, and user-information handling.

Through this module, I gained practical knowledge of **C++ programming and Object-Oriented Programming concepts**. I understood how classes, objects, functions, data validation, and modular programming can be used to develop a structured software system.

I also learned the importance of secure user access and reliable authentication in software applications. Testing the module helped me understand how to handle valid and invalid login attempts, incorrect passwords, duplicate user information, and different user-management situations.

Overall, my contribution to Module 1 helped establish the user-access foundation of the Print Ease system and provided valuable experience in designing a practical C++-based software solution.

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APPENDICES

Appendix A – Detailed Calculations

This appendix contains the essential calculations used in the Print Ease Smart Campus Printing System, including:

A.1 Print Cost Calculation

The estimated printing cost is calculated based on:

- Number of pages
- Number of copies
- Selected printing options

General calculation:

Total Cost = Number of Pages × Number of Copies × Cost per Page

Additional charges may be included depending on the selected printing options.

A.2 Priority Scheduling Calculation

The print-job scheduling mechanism considers factors such as:

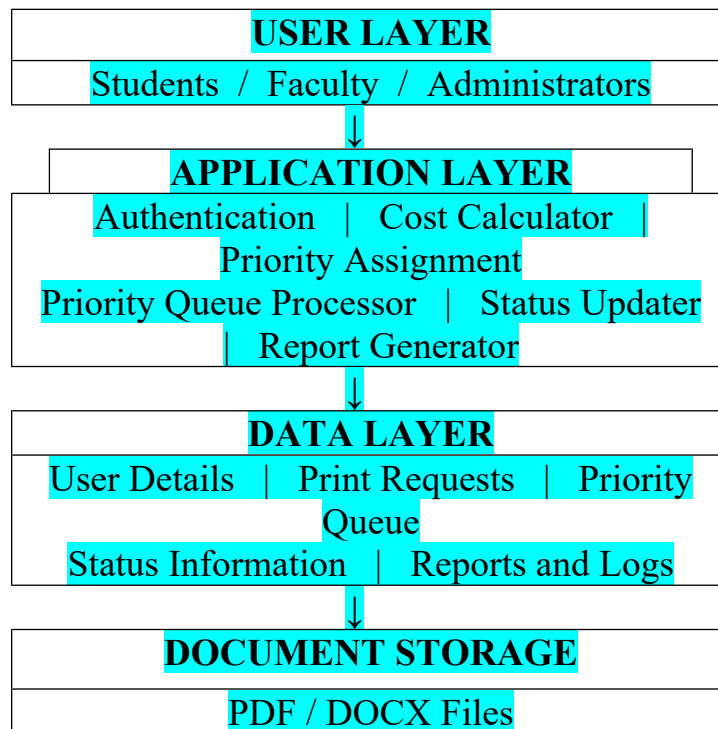
- Base priority
- Waiting time
- Urgency
- User category

The effective priority is updated to improve fairness and reduce the possibility of starvation among waiting jobs.

Appendix B – Engineering Drawings / Schematics

This appendix contains the essential engineering diagrams and system schematics of the Print Ease system.

B.1 System Architecture



Appendix C – Source Code / Code Repository Information

Only essential source-code excerpts are included in this report. The complete implementation is maintained separately as part of the project files.

C.1 Core Technologies Used

- Backend: C++17
- Frontend: HTML5, CSS3, JavaScript ES6
- Development Environment: Visual Studio Code
- Communication: WinSock2
- Core Scheduling Structure: Binary Max-Heap Priority Queue

C.2 Source Code Components

The complete source code contains modules for:

- User registration and login
- Print-request management
- Priority-based scheduling
- Binary Max-Heap implementation
- Adaptive priority scheduling
- Cost calculation
- Print-status tracking
- Administrative functions
- Frontend and backend integration

The complete source code may be maintained in the approved project repository or submitted separately as required by the institution.

Appendix D – Datasheets

As Print Ease is a software-based project, no specialised hardware components or physical sensors are used.

The technical documentation and software resources applicable to the project include:

- C++17 programming documentation
- WinSock2 API documentation
- HTML5, CSS3, and JavaScript documentation
- Development environment documentation

Therefore, no traditional electronic or hardware datasheets are required for the current project implementation.

Appendix E – Experimental / Raw Data

This appendix contains the raw test data collected during system testing.

The test data may include:

- User authentication test results
- Print-request submission results
- Priority Queue ordering results
- Adaptive priority scheduling observations

A sample raw-data format is shown below:

Test ID	Function Tested	Expected Result	Actual Result	Status
T01	User Login	Valid user should log in successfully	Successful login	Pass
T02	Invalid Login	Invalid credentials should be rejected	Access rejected	Pass
T03	Priority Scheduling	High-priority job selected first	Correct order obtained	Pass
T04	Cost Calculation	Correct estimated cost displayed	Correct result obtained	Pass
T05	Status Tracking	Correct status progression displayed	Pending → Printing → Completed	Pass

Appendix F – Ethics / Institutional Approval

The project is a software-based academic project and does not involve clinical experiments, hazardous materials, human medical data, or sensitive personal data collection.

The system should follow the following ethical considerations:

- User information should be handled responsibly.
- Uploaded documents should not be accessed without authorisation.
- Priority scheduling should remain fair to all users.
- Low-priority jobs should not be indefinitely delayed.
- Authentication and access control should be implemented appropriately.

Formal institutional approval may be included where required by the institution.

Appendix G – Project Meeting / Progress Records

This appendix records the major stages of project development and team discussions.

Stage	Activity	Outcome
Planning	Problem identification and project discussion	Smart campus printing problem identified
Requirement Analysis	System requirements discussed	Functional and non-functional requirements defined
Design	Architecture and data structures selected	Binary Max-Heap Priority Queue selected
Development	Frontend and backend implementation	Core system modules developed
Integration	Frontend–backend communication	System components integrated
Testing	Functional verification	Major features tested
Documentation	Report preparation	Final project documentation prepared

Appendix H – User / Industry Feedback

Where applicable, feedback from potential users can be collected from:

- Students
- Faculty members
- Printing staff

The feedback may focus on:

- Ease of using the website
- Clarity of the print-request process
- Usefulness of priority-based scheduling
- Accuracy of cost estimation
- Effectiveness of print-status tracking

A suggested feedback format is:

Feedback Parameter	User Response	Suggested Improvement
Ease of Use	To be collected	Improve interface where required
Print Request Process	To be collected	Simplify submission process
Status Tracking	To be collected	Improve status visibility
Priority Scheduling	To be collected	Review fairness rules

Appendix I – Publications / Patents / Competitions / Product Outcomes

At the current stage, the project outcome is an academic software prototype named Print Ease: Smart Campus Printing System.

Potential future outcomes may include:

- Participation in student project competitions.
- Further development into a deployable campus application.
- Presentation of the project at academic events.
- Development of additional features such as multi-printer support and online payment.

No publication, patent, or commercial product outcome is claimed unless officially achieved and documented.

Appendix J – Individual Contribution Evidence

The Print Ease project was developed as a team project. Each team member contributed to different stages of development, implementation, testing, and documentation.

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Contribution
Ragul E (192571032)	Project coordination and core website development	Developed the main Print Ease website and core system features	Tested website functionality and user interactions	Introduction, Objectives, and System Design	25%
Manoj M (192511060)	Frontend design and user interface development	Developed the website layout, navigation, forms, and user interface	Tested page navigation and usability	Literature Review and Methodology	25%
DEVA.J (192421258)	Print management and scheduling	Developed or supported priority scheduling and print management features	Tested scheduling logic and performance	Implementation and Testing	25%
JAYAKRISHNA.V (192421259)	System integration and documentation	Integrated system components and supported functionality	Performed final testing and error analysis	Results, Conclusion, and References	25%
Total	Complete Project Development	—	—	—	100%